

Topic 1: Data Representation – Exercise Sheet

[Solutions]

Q#1: We have discussed Binary, Denary (aka Decimal), and Hexadecimal number systems in class. Complete the following table, doing the necessary conversions (without using a calculator and with a little-bit of self learning using the notes given below).

Decimal	Binary	Octal	Hexadecimal
175	10101111	257	AF
204	11001100	314	CC
503	111110111	767	1F7
3995	111110011011	7633	F9B

Notes: octal number system is a base-8 numeral system that uses digits from 0 to 7. It is commonly used in computing as a more compact way to represent binary numbers, with each octal digit corresponding to a group of three binary digits. Octal is less commonly used today but was historically useful in systems like UNIX file permissions and in early computing architectures. For example, the binary number 110010 can be grouped as 110 010 and represented as 62 in octal. For more details, please read *Chapter #2, Section 2.2.4* of *Forouzan*.

Q#2: In a positional number system with base b , the largest integer number that can be represented using k digits is $b^k - 1$. Find the largest number in each of the following systems with **six** digits.

- (a) Binary $2^6 - 1 = 63$
- (b) Decimal $10^6 - 1 = 999999$
- (c) Hexadecimal $16^6 - 1 = 16777215$
- (d) Octal $8^6 - 1 = 262143$